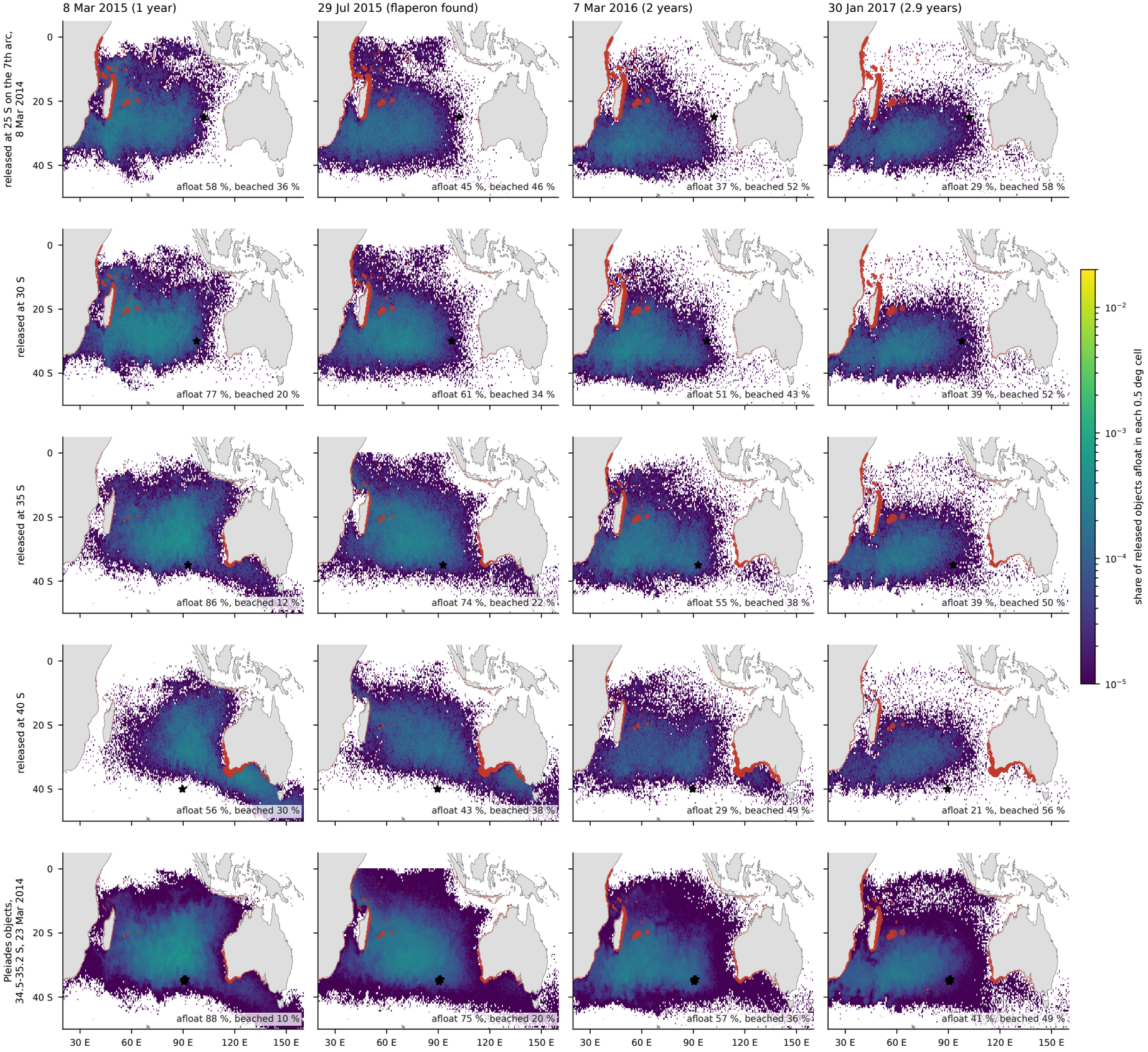
Where floating debris goes, March 2014 to January 2017 (two ocean models, equal weight)

Colour: objects still afloat. Red dots: where objects have come ashore so far (dot area = share). Star: release point.

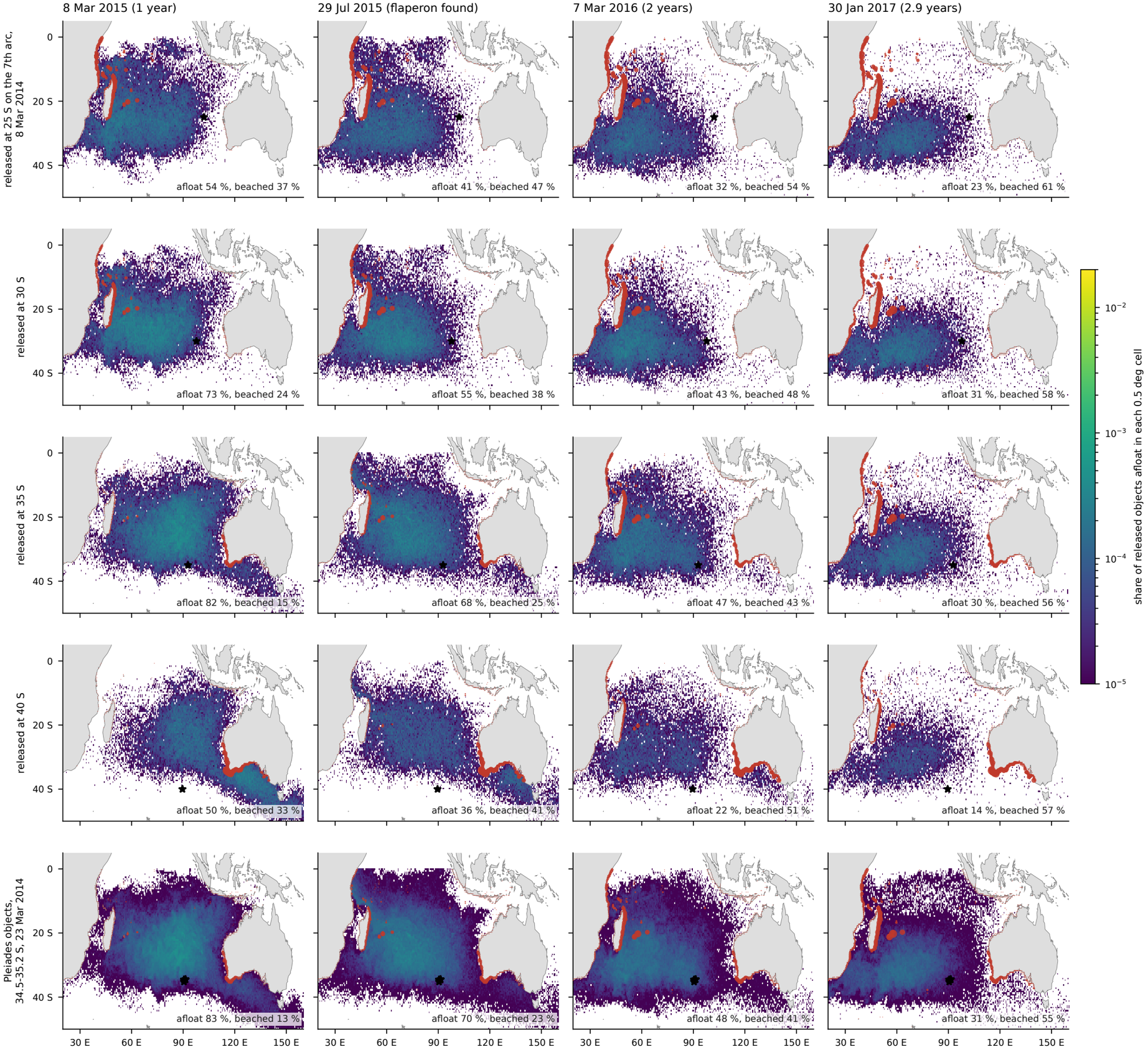


Plain English. Each start point releases 100,000 floating objects of three types (flaperon-like, low-exposure exterior parts, high-windage interior parts), in four ocean realisations each. The objects drift with the ocean current and with part of the wind, plus small random eddies. They stop when they reach a coast and do not float off again. Debris is released at the impact point on 8 Mar 2014; the Pleiades objects are released where and when they were imaged (six object groups pooled), which assumes they are aircraft debris. The model tracks the region 15-160 E, 50 S to the equator; objects that leave it are no longer shown. Two ocean models (GLORYS12 and GlobCurrent, with ERA5 wind) have equal weight.

Technical. Untracked stand-in harness `snapshots` on drift d51d80a (OceanSetup::run_tracks_ep; class_responses; diffusivity_prior; ocean error per env seed), configs aus160-<model>.toml (15-160 E, GSHHG 2.3.7 coast, snap 25 km, refloat off, 6 h step, K log-uniform 100-1000 m2/s, env seeds 1-4, 8,334 particles per class, no importance splitting). GlobCurrent windage product-relative U(-0.75,-0.60) % of U10. Release 2014-03-08 00:19:37 UTC at 7th-arc longitudes 102.10/97.78/92.83/89.58 E (median of run J impacts within 0.1 deg); Pleiades rating-5 targets (targets-3km.csv) at 04:24-04:28 UTC 23 Mar 2014. Beachings in the window count from the first episode. Share = count / released, equal weight per ocean model.

Where floating debris goes, March 2014 to January 2017 (GLORYS12 ocean model only)

Colour: objects still afloat. Red dots: where objects have come ashore so far (dot area = share). Star: release point.

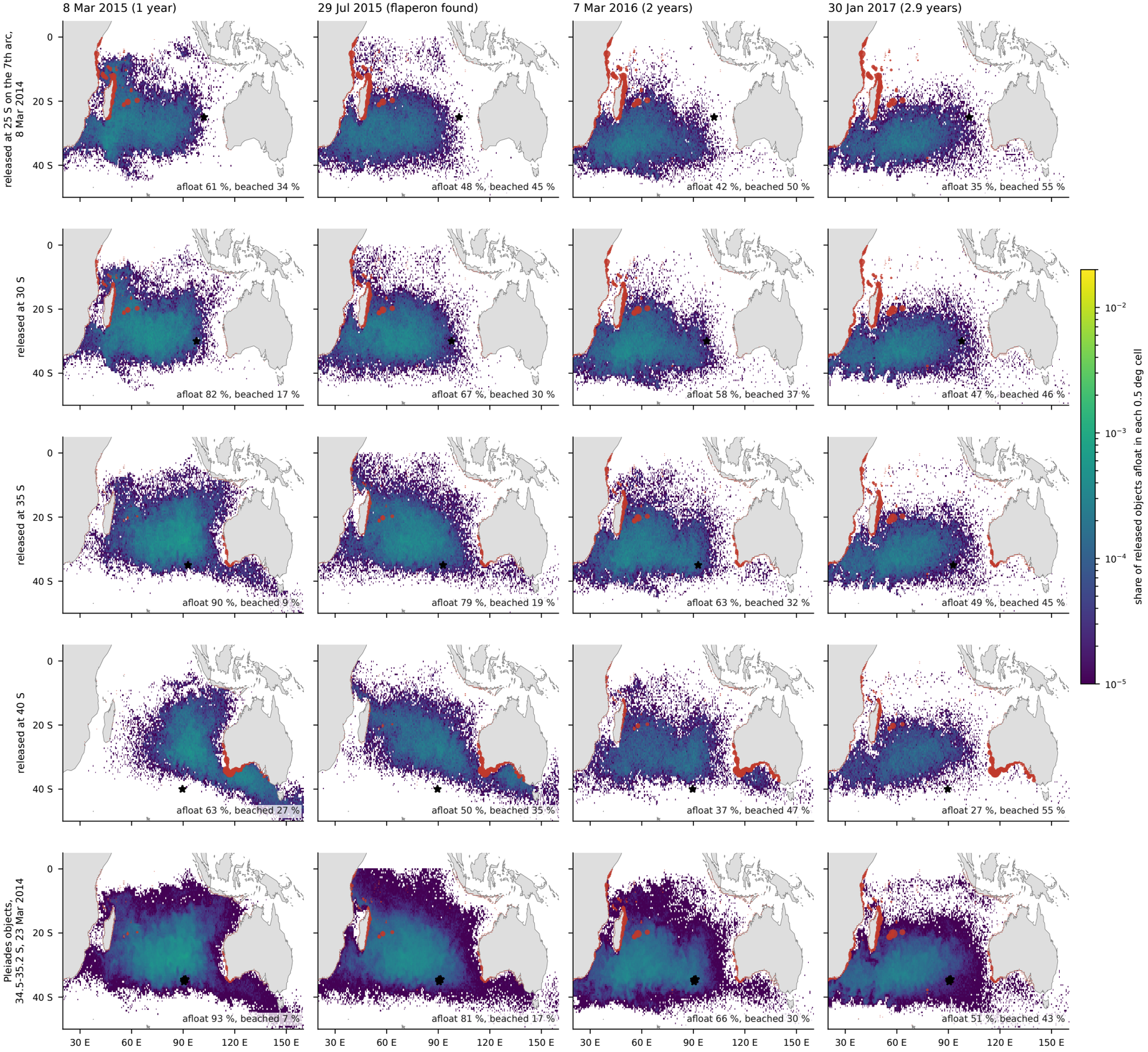


Plain English. Each start point releases 100,000 floating objects of three types (flaperon-like, low-exposure exterior parts, high-windage interior parts), in four ocean realisations each. The objects drift with the ocean current and with part of the wind, plus small random eddies. They stop when they reach a coast and do not float off again. Debris is released at the impact point on 8 Mar 2014; the Pleiades objects are released where and when they were imaged (six object groups pooled), which assumes they are aircraft debris. The model tracks the region 15-160 E, 50 S to the equator; objects that leave it are no longer shown. Ocean model: GLORYS12 with ERA5 wind.

Technical. Untracked stand-in harness `snapshots` on drift d51d80a (OceanSetup::run_tracks_ep; class_responses; diffusivity_prior; ocean error per env seed), configs aus160-<model>.toml (15-160 E, GSHHG 2.3.7 coast, snap 25 km, refloat off, 6 h step, K log-uniform 100-1000 m2/s, env seeds 1-4, 8,334 particles per class, no importance splitting). GlobCurrent windage product-relative U(-0.75,-0.60) % of U10. Release 2014-03-08 00:19:37 UTC at 7th-arc longitudes 102.10/97.78/92.83/89.58 E (median of run J impacts within 0.1 deg); Pleiades rating-5 targets (targets-3km.csv) at 04:24-04:28 UTC 23 Mar 2014. Beachings in the window count from the first episode. Share = count / released, equal weight per ocean model.

Where floating debris goes, March 2014 to January 2017 (GlobCurrent ocean model only)

Colour: objects still afloat. Red dots: where objects have come ashore so far (dot area = share). Star: release point.



Plain English. Each start point releases 100,000 floating objects of three types (flaperon-like, low-exposure exterior parts, high-windage interior parts), in four ocean realisations each. The objects drift with the ocean current and with part of the wind, plus small random eddies. They stop when they reach a coast and do not float off again. Debris is released at the impact point on 8 Mar 2014; the Pleiades objects are released where and when they were imaged (six object groups pooled), which assumes they are aircraft debris. The model tracks the region 15-160 E, 50 S to the equator; objects that leave it are no longer shown. Ocean model: GlobCurrent with ERA5 wind.

Technical. Untracked stand-in harness `snapshots` on drift d51d80a (OceanSetup::run_tracks_ep; class_responses; diffusivity_prior; ocean error per env seed), configs aus160-<model>.toml (15-160 E, GSHHG 2.3.7 coast, snap 25 km, refloat off, 6 h step, K log-uniform 100-1000 m2/s, env seeds 1-4, 8,334 particles per class, no importance splitting). GlobCurrent windage product-relative U(-0.75,-0.60) % of U10. Release 2014-03-08 00:19:37 UTC at 7th-arc longitudes 102.10/97.78/92.83/89.58 E (median of run J impacts within 0.1 deg); Pleiades rating-5 targets (targets-3km.csv) at 04:24-04:28 UTC 23 Mar 2014. Beachings in the window count from the first episode. Share = count / released, equal weight per ocean model.